

Image Segmentation:

- We want packages to work with tif files with different channels (red, blue, green, etc.). We also want packages that work with IF images

OpenCV:

Can work with .tiff files, but do not work well with immunofluorescence images – built for simple images with less color channels.

Scikit-learn (scikit-image):

Support multi-channel tif files. Skimage.io.imread

Works well with IF image, can do segmentation

Napari: (Graphical Interface)

<https://github.com/napari/napari/>

Segmentation – not built into basic napari package (only include color assignment, label filtering, etc.). The actual segmentation is done via plugins:

Example 1: <https://github.com/haesleinhuepf/napari-segment-blobs-and-things-with-membranes>

Based on scikit-image – segment nuclei and cells based on fluorescence images with high intensity in nuclei.

End result: label bright objects such as nuclei in an intensity image.

Example 2: <https://github.com/stardist/stardist-napari/>

Stardist – deep learning based segmentation method to detect object with star shapes (compatible with nuclei and cells). Originally developed for fluorescence microscopy images.

LIMITATION: assume star-convex shape nuclei

Example 3: <https://github.com/MouseLand/cellpose-napari>

Cellpose – not a lot documentation found online – uses deep learning?

Example 4: <https://github.com/MIC-DKFZ/napari-sam>

Sam – is not just limited to IF image – prompt based